

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (21 marks – approximately 38 minutes)

(a) The following functions represent the market for wheat in Country A:

$$Q(s) = 150 + 30P$$

$$Q(d) = 500 - 40P$$

where P is the price per bushel of wheat.

Required:

- (i) Solve for the market equilibrium. (3 marks)
 - (ii) If the government controls the price of wheat at HK\$4 per bushel, calculate the amount of excess demand or excess supply at a price of HK\$4. (4 marks)
 - (iii) State the TWO major forms of price controls. (2 marks)
 - (iv) State any ONE market-failure related cause that explains the type of price control adopted by government in part (a)(ii) and the market practice likely to emerge as a result. (2 marks)
- (b) Analyse the change in the equilibrium price and quantity in a competitive automotive market in each of the following cases.

Note: No graphical support is required.

Required:

- (i) The terrorist attacks in some petroleum refinery facilities in the Middle East triggered the increase in price of gasoline. (5 marks)
- (ii) The market price of steel has increased significantly. (5 marks)

Question 2 (16 marks – approximately 29 minutes)

International trade has long been one of the key engines for national and global economic growth. In recent years, nationalism and protectionism have been gaining more momentum even among developed economies.

Required:

- (a) Define gains from international trade and state any TWO of its sources. (3 marks)
- (b) Compare tariff and quota as trade protectionist measures. (4 marks)
- (c) Hong Kong has long been a highly externally-oriented economy that relies on international trade and finance to flourish.
- (i) Describe a commonly-used measure for an economy's "degree of trade openness". (2 marks)
- (ii) Based on the measure in part (c)(i), compare the degree of trade openness between Hong Kong and the US. (1 mark)
- (iii) State the TWO fundamental types of exchange rate systems. (2 marks)
- (iv) Describe the exchange rate system adopted by Hong Kong in relation to the answer to part (c)(iii). (1 mark)
- (v) State ONE advantage and ONE disadvantage of Hong Kong's exchange rate system to the territory. (3 marks)

Question 3 (19 marks – approximately 34 minutes)

Country X consumes a lot of chocolate bars. In 2019, the country consumed 100 Venus chocolate bars for HK\$9 each and 200 Bradbury hazelnut chocolate bars for HK\$10 each. In 2020, the country consumed 130 Venus chocolate bars for HK\$10 each and 220 packs of Bradbury hazelnut chocolate bars for HK\$12 each.

Required:

- (a) Year 2019 is the base year. Compute the following price indices for years 2019 and 2020:
- GDP Deflator; and
 - Consumer Price Index.
- (8 marks)
- (b) Based on the TWO indices above, compute the percentage change in price between 2019 and 2020. Analyse the differences between the TWO answers.
- (7 marks)
- (c) From an economist's viewpoint, explain which of the above indices is considered a better measure of the cost of Country X's chocolate bar consumption over time. Assume that there are rival brands of chocolate bar in the market.
- (4 marks)

Question 4 (24 marks – approximately 43 minutes)

(a) Statistics, as a discipline, is typically divided into two categories: descriptive and inferential.

(i) Describe inferential statistics and random sampling methods.
(4 marks)

(ii) List TWO advantages and TWO disadvantages of random sampling method in statistical inference.
(2 marks)

(iii) State and describe any TWO random sampling methods.
(6 marks)

(b) State and describe the two defining characteristics of public goods.
(4 marks)

(c) Jocelyn is an economics lecturer. She has ranked her three options of what she can do during the summer in the following order:

- (1) work as a consultant full-time and earn HK\$35,000;
- (2) teach summer school for the first eight weeks of summer and earn HK\$20,000 and spend the rest of the summer travelling; and
- (3) write textbook supplements for a publisher full-time and earn HK\$30,000.

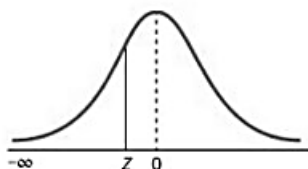
Required:

Explain the opportunity cost of Jocelyn's choice.
(8 marks)

* * * END OF EXAMINATION PAPER * * *

APPENDIX

The Cumulative Standardised Normal Distribution



Entry represents area under the cumulative standardised normal distribution from $-\infty$ to Z										
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641